Digital Electronics (ECT-033) - Unit 4

Unit 4 - Logic Families & Semiconductor Memories - notes in easy English, matched to the UTU syllabus.

Study Materials by Anas - study-materials-by-anas.vercel.app

What this unit covers: logic families (how real gates are built) and semiconductor memories.

Key ideas: Logic families: **TTL** (transistor-transistor logic - fast, bipolar), **ECL** (emitter-coupled - fastest, high power), **MOS/CMOS** (lowest power, high density, slower than TTL). Compare them on speed, power dissipation, fan-in/fan-out, noise margin - a comparison table question is common. Key parameters: **propagation delay**, **power dissipation**, **fan-out** (how many inputs one output can drive), **noise margin**. Memories: **RAM** (volatile - SRAM is fast flip-flop based, DRAM is capacitor based, denser, needs refresh), **ROM** (non-volatile - PROM, EPROM, EEPROM differ in how they are programmed/erased). Memory capacity questions: a chip with m address lines and n data lines stores 2^k x n bits.

Exam pointers: TTL vs CMOS comparison and RAM vs ROM (with SRAM/DRAM and PROM/EPROM/EEPROM sub-types) are the classic 5-7 mark theory questions. Practise one memory-size numerical (address lines → capacity).